

Orthogonal Projections

Lecture 2

We start assuming a linear model,

$$\underbrace{y_i}_{1 \times 1} = \underbrace{x'_i}_{1 \times p} \underbrace{\beta}_{p \times 1} + \underbrace{u_i}_{1 \times 1} \quad \text{for } i = 1, 2, \dots, n, \quad (1)$$

where y is the response variable for subject i and x_i is a vector of p independent variables that includes an intercept. The model can be written in a more convenient matrix notation as

$$\underbrace{y}_{n \times 1} = \underbrace{\underbrace{X}_{n \times p} \underbrace{\beta}_{p \times 1}}_{n \times 1} + \underbrace{u}_{n \times 1} \quad (2)$$

where y is now defined as a $n \times 1$ vector, the matrix X is $n \times p$, the vector of parameters β is $p \times 1$, and the random vector u is $n \times 1$. In other words,

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p-1} \\ 1 & x_{21} & x_{22} & \dots & x_{2p-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np-1} \end{bmatrix}; \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}; \quad u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}.$$

It follows that

$$\sum_{i=1}^n x_i x'_i = X'X,$$

$$\sum_{i=1}^n x_i y_i = X'y.$$

1 Estimation

We can estimate the parameter β using the Method of Moments or alternatively the traditional method called Ordinary Least Squares (OLS). The derivation of the OLS estimator is

based on minimizing the sum of squared errors

$$||u||^2 = ||y - X\beta||^2$$

where $||\cdot||$ is the Euclidean norm defined as $(z'z)^{1/2} = (\sum z_i^2)^{1/2}$ for a generic vector z . Therefore, the objective function is

$$\begin{aligned} \min_{\beta} ||y - X\beta||^2 &= \min_{\beta} (y - X\beta)'(y - X\beta) = \min_{\beta} \{(y' - \beta'X')(y - X\beta)\} \\ &= \min_{\beta} \{y'y - \beta'X'y - y'X\beta + \beta'X'X\beta\} \\ &= \min_{\beta} \{y'y - 2\beta'X'y + \beta'X'X\beta\} \end{aligned}$$

because $\beta'X'y = y'X\beta$. The first order condition is

$$-2X'y + 2X'X\beta = 0,$$

which implies that $X'(y - X\beta) = 0$, and consequently, under Assumptions A1, A2 and A3:

$$\hat{\beta} = (X'X)^{-1}X'y. \quad (3)$$

Alternatively, we define the estimator as

$$\hat{\beta} = \arg \min ||y - X\beta||^2 \quad (4)$$

The OLS estimator $\hat{\beta}$ gives the closest vector $X\hat{\beta}$, in terms of the Euclidean norm, to y among all vectors that lie on the space spanned by the vector X .

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We can also obtain the variance of the estimator under homocedasticity and independent errors (Assumptions A4 and A5). Note that,

$$\begin{aligned} V(\hat{\beta}|X) &= V((X'X)^{-1}X'y|X) = V((X'X)^{-1}X'(X\beta + u)|X) \\ &= (X'X)^{-1}X'V(u|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\sigma_u^2 I_N X(X'X)^{-1} \\ &= \sigma_u^2 (X'X)^{-1}, \end{aligned}$$

under the assumption that the variance of the error term is “spherical” i.e. $V(u|X) = \sigma_u^2 I$. The variance depends on σ_u^2 , which is unknown and needs to be estimated. As in Lecture 1, it is estimated using $\hat{u}'\hat{u}/n - p$.

Note also that the homocedasticity assumption simplifies the expression of the variance

of OLS.

2 Finite Sample Properties

The OLS estimator $\hat{\beta}$ is unbiased when the independent variables are independent of the error term (Assumption A2),

$$\begin{aligned} E(\hat{\beta}|X) &= E((X'X)^{-1}X'y|X) \\ &= \beta + E((X'X)^{-1}X'u|X) \\ &= \beta + (X'X)^{-1}X'E(u|X) \\ &= \beta \end{aligned}$$

By the law of iterated expectations, we also have that

$$E(\hat{\beta}) = E(E(\hat{\beta}|X)) = \beta$$

The OLS estimator is unbiased and linear, which means that we can write $\hat{\beta}$ as a linear function of the response variable y e.g.

$$\hat{\beta} = Ay$$

where $A = (X'X)^{-1}X'$. Now, consider another unbiased linear estimator,

$$\tilde{\beta} = (A + B)y$$

Therefore, it must be the case that $E\tilde{\beta} = \beta$ because $BX = 0$. Also, the variance of the estimator $\tilde{\beta}$ is,

$$\begin{aligned} V(\tilde{\beta}|X) &= E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] \\ &= (A + B)Euu'(A + B)' \\ &= \sigma^2(X'X)^{-1} + \sigma^2BB' \\ &= V(\hat{\beta}|X) + \sigma^2BB' \end{aligned}$$

The last equation suggests that the variance of the estimator $\tilde{\beta}$ is larger than the variance of the estimator $\hat{\beta}$. This can be stated in the following important theorem,

Theorem 1. (Gauss-Markov) If $E(u|X) = 0$ and $E(uu'|X) = \sigma^2I$ in a linear regression model, then the OLS estimator is efficient e.g. $V(\tilde{\beta}|X) \geq V(\hat{\beta}|X)$ or

$$V(\tilde{\beta}|X) - V(\hat{\beta}|X) \text{ is positive semi-definite}$$

Note that the theorem is for conditional variances. Can we claim that $V(\tilde{\beta}) \geq V(\hat{\beta})$ also? We need a bit more work ... Consider first the definition

$$V(\hat{\beta}) = E(V(\hat{\beta}|X)) + V(E(\hat{\beta}|X)).$$

First, note that the second term is equal to zero because the estimator is unbiased, and therefore, $V(\beta) = 0$. Thus,

$$V(\hat{\beta}) = E(V(\hat{\beta}|X)) = \sigma^2 E((X'X)^{-1}).$$

Similarly,

$$V(\tilde{\beta}) = E(V(\tilde{\beta}|X)) = \sigma^2 [E((X'X)^{-1}) + E(BB')].$$

Therefore, we conclude that

$$V(\tilde{\beta}) \geq V(\hat{\beta}),$$

because $E(BB')$ is positive semi-definite.

3 Projections

Define the projection matrix

$$P = X(X'X)^{-1}X'.$$

The i th diagonal element of P is labeled leverage point and is defined as $h_i = x_i'(X'X)^{-1}x_i$. The matrix P is symmetric and idempotent, and its trace is equal to the number of independent variables. Symmetry implies,

$$P' = (X(X'X)^{-1}X')' = (X')'((X'X)^{-1})'(X)' = X(X'X)^{-1}X' = P.$$

Idempotency implies,

$$PP = X(X'X)^{-1}X'X(X'X)^{-1}X' = X(X'X)^{-1}X' = P.$$

Lastly,

$$\text{tr}(P) = \text{tr}(X(X'X)^{-1}X') = \text{tr}(X'X(X'X)^{-1}) = \text{tr}(I_p) = p.$$

The vector of fitted values, defined as

$$\hat{y} = X\hat{\beta},$$

represents the orthogonal projection of y onto the column space of X . Note that,

$$\hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = Py$$

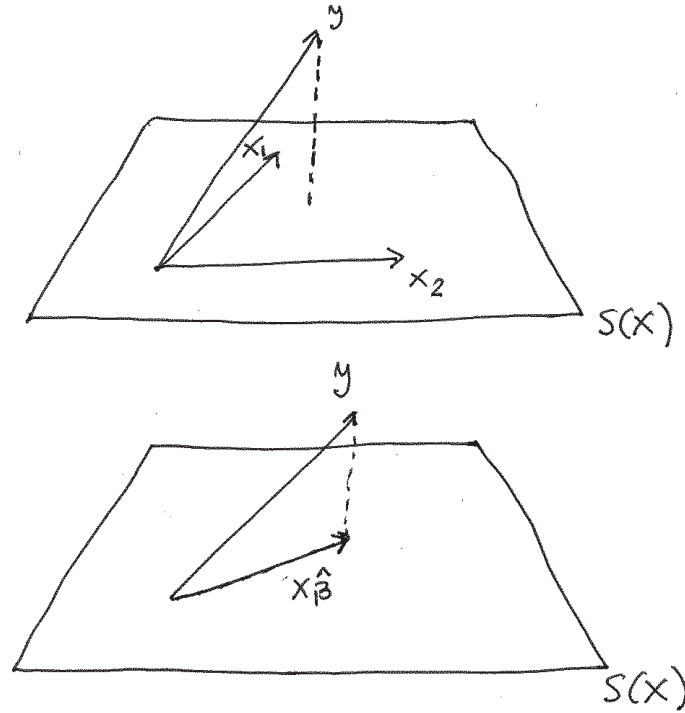


Figure 1: OLS fitting and projections

where the matrix P is the projection matrix. On the other hand, the residual vector

$$\hat{u} = y - \hat{y} = (I - P)y = My,$$

where $M := I - P$, which is a matrix that produces ‘residuals’. Note that we can also write the vector of residuals as

$$\hat{u} = My = M(X\beta + u) = MX\beta + Mu = Mu$$

because $PX = X$. This implies that the residuals from the regression and the independent variables are orthogonal,

$$X'\hat{u} = X'Mu = X'(I - P)u = (X' - X'P)u = 0.$$

Remark 1. The basic geometry of OLS can be seen in Figure 1. Let y and $X = [X_1 : X_2]$ be vectors, and define the space spanned by the columns of X as:

$$S(X) := \{z \in R^n | z = Xc, \quad c \in R^p\}$$

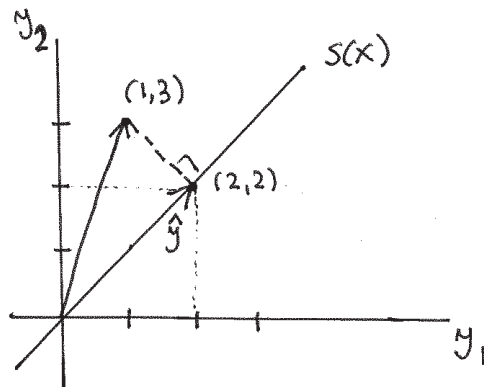


Figure 2: OLS fitting with two observations

The OLS fitting gives the closest vector to y in the space spanned by the X s.

Example 1. To illustrate the basic geometry of OLS, we present a simple example (Figure 2). Let $y = (1, 3)'$ and $X = (1, 1)'$ be vectors, and consider the regression model $y_i = \beta X_i + u_i$. The OLS estimator is

$$\arg \min \{(y_1 - \beta)^2 + (y_2 - \beta)^2\}$$

and the associated first order condition,

$$-2(y_1 - \beta) - 2(y_2 - \beta) = 0, \Rightarrow \bar{y} = \frac{1}{2} \sum y_i = 2.$$

Therefore, we obtain $\hat{y} = \hat{\beta}X = (2, 2)'$ and $\hat{u} = y - \hat{y} = (-1, 1)'$ with $X'\hat{u} = 0$. As it is clear in the figure, it is also interesting to obtain the length of the following vectors using the Euclidean norm:

$$\begin{aligned} \|y\| &= \sqrt{1^2 + 3^2} = \sqrt{10}, \quad \|\hat{y}\| = \sqrt{2^2 + 2^2} = \sqrt{8}, \\ \|u\| &= \|y - \hat{y}\| = \sqrt{(y_1 - 2)^2 + (y_2 - 2)^2} = \sqrt{2}. \end{aligned}$$

4 Partitioned Fit (Frish-Waugh-Lovell Theorem)

Consider a partitioned regression model,

$$y = X\beta + Z\gamma + u, \tag{5}$$

and the corresponding estimator $\hat{\beta}$. Moreover, consider the alternative regression model,

$$\tilde{y} = \tilde{X}\beta + \tilde{u}, \quad (6)$$

with the corresponding estimator $\tilde{\beta}$, with $\tilde{z} = M_Z z$ for a generic vector z .

Theorem 2. *The estimators $\hat{\beta}$ and $\tilde{\beta}$ are numerically identical.*

Proof: Standard OLS formulae in model (5) gives the following normal equations,

$$\begin{aligned} X'y &= X'X\hat{\beta} + X'Z\hat{\gamma} \\ Z'y &= Z'X\hat{\beta} + Z'Z\hat{\gamma} \end{aligned}$$

Solving for $\hat{\gamma}$ in the second equation, and replacing in the first equation, we obtain

$$X'y = X'X\hat{\beta} + X'Z((Z'Z)^{-1}Z'y - (Z'Z)^{-1}Z'X\hat{\beta}),$$

suggesting that

$$X'(I - Z(Z'Z)^{-1}Z')y = X'(I - Z(Z'Z)^{-1}Z')X\hat{\beta}$$

Therefore,

$$\hat{\beta} = (X'M_Z X)^{-1}X'M_Z y = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y} = \tilde{\beta}$$

■

In addition to the point estimates, the residuals are identical. To see this consider,

$$y = \hat{y} + \hat{u} = X\hat{\beta} + Z\hat{\gamma} + \hat{u}$$

and multiply by M_Z ,

$$M_Z y = M_Z X\hat{\beta} + M_Z Z\hat{\gamma} + M_Z \hat{u} \Rightarrow \tilde{y} = \tilde{X}\tilde{\beta} + M_Z \hat{u},$$

but $M_Z \hat{u} = \hat{u}$ because $P_Z \hat{u} = 0$.

Another way is seeing the this result is considering the following argument. Recall that

$$y = \hat{y} + \hat{u} = Py + My = X\hat{\beta} + Z\hat{\gamma} + My \quad (7)$$

Note that M is a projection matrix that uses $[X:Z]$. Applying partitioned formulae, one can show that $M = I - P_{XZ} = I - (P_Z + M_Z X(X'M_Z X)^{-1}X'M_Z)$.

If we multiply (7) by M_Z , we have

$$M_Z y = M_Z X\hat{\beta} + M_Z Z\hat{\gamma} + M_Z My \quad (8)$$

Therefore, we need to show that $M_Z M = M$ or, in words, that the last term in equations

(7) and (8) are equal. Note that

$$\begin{aligned}
M_Z M &= (I - P_Z)(I - P_{XZ}) \\
&= I - P_Z - P_{XZ} + P_Z P_{XZ} \\
&= I - P_Z - P_{XZ} + P_Z(P_Z + M_Z X(X' M_Z X)^{-1} X' M_Z) \\
&= I - P_Z - P_{XZ} + P_Z P_Z + P_Z M_Z X(X' M_Z X)^{-1} X' M_Z \\
&= I - P_Z - P_{XZ} + P_Z + 0 \\
&= I - P_{XZ} = M,
\end{aligned}$$

which gives the results since $My = Mu = \hat{u}$.

Note too that in the case of $Z'X = 0$, the “off-diagonal” elements of the inverse matrix in the partitioned formula are zero, then $M = I - P_{XZ} = I - (P_Z + P_X)$. Again, we have that $P_Z(P_Z + P_X) = P_Z$, and the result continues to hold.

The theorem implies that the OLS estimates of the slope parameter β can be obtained as follows:

1. Regress y on Z and obtain the residuals $\tilde{y}$
2. Regress X on Z and obtain the residuals $\tilde{X}$
3. Regress $\tilde{y}$ on $\tilde{X}$, and obtain the OLS estimate $\hat{\beta}$.

Example 2. Consider a simple bivariate model,

$$y = \gamma + \beta x_i + u_i$$

where in this case, $X = x_i$ and $Z = \mathbf{1}$. Before turning to the application of FWL theorem, we can use the method of moments to derive the slope estimator. Our model assumptions were

$$\begin{aligned}
E(u) &= E(y - \gamma - \beta x) = 0 \\
\text{Cov}(x, u) &= E(xu) - E(x)E(u) = E(xu) = E(x(y - \gamma - \beta x)) = 0
\end{aligned}$$

These two equations are restrictions imposed to the data about the joint probability distribution of x and y . There are two unknowns and two equations, which suggests that we can solve for γ and β . The sample counterpart of these equations are

$$\frac{1}{n} \sum_{i=1}^n (y_i - \gamma - \beta x_i) = 0 \quad (9)$$

$$\frac{1}{n} \sum_{i=1}^n (x_i(y_i - \gamma - \beta x_i)) = 0 \quad (10)$$

From equation (6), we obtain $\hat{\gamma} = \bar{y} - \hat{\beta}\bar{x}$, and replacing in (8), we obtain

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta} \sum_{i=1}^n (x_i - \bar{x})^2$$

which implies that the estimator for the parameter β is

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (11)$$

Following the FWL theorem, we first have to obtain the residuals in the regression,

$$x_i = \alpha + \epsilon_i \Rightarrow \tilde{x}_i = x_i - \hat{\alpha} = x_i - \bar{x}$$

Similarly,

$$y_i = \delta + v_i \Rightarrow \tilde{y}_i = y_i - \hat{\delta} = y_i - \bar{y}$$

Finally, we have to regress the residuals of the response variable on the residuals of the variable of interest,

$$\tilde{y}_i = \pi \tilde{x}_i + w_i$$

giving,

$$\hat{\pi} = \frac{\sum \tilde{x}_i \tilde{y}_i}{\sum \tilde{x}_i^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \hat{\beta} \quad (12)$$

Let's check now that the residuals in the two equations are equal, so the standard errors of $\hat{\beta}$ and $\hat{\pi}$ are equal. Again, by definition,

$$y_i = \hat{\gamma} + \hat{\beta}x_i + \hat{u}_i, \quad (13)$$

and

$$\tilde{y}_i = \hat{\pi} \tilde{x}_i + \hat{w}_i. \quad (14)$$

We can write the last equation as,

$$y_i - \bar{y} = \hat{\pi}(x_i - \bar{x}) + \hat{w}_i = \hat{\beta}(x_i - \bar{x}) + \hat{w}_i$$

and therefore,

$$\begin{aligned} y_i &= (\bar{y} - \hat{\beta}\bar{x}) + \hat{\beta}x_i + \hat{w}_i, \\ &= \hat{\gamma} + \hat{\beta}x_i + \hat{w}_i, \end{aligned}$$

using the definition of the intercept $\hat{\gamma}$. (Recall that we discuss this in Lecture 1). Notice then that the fitted values are equal, and therefore, the residual in equation (13) is equal to

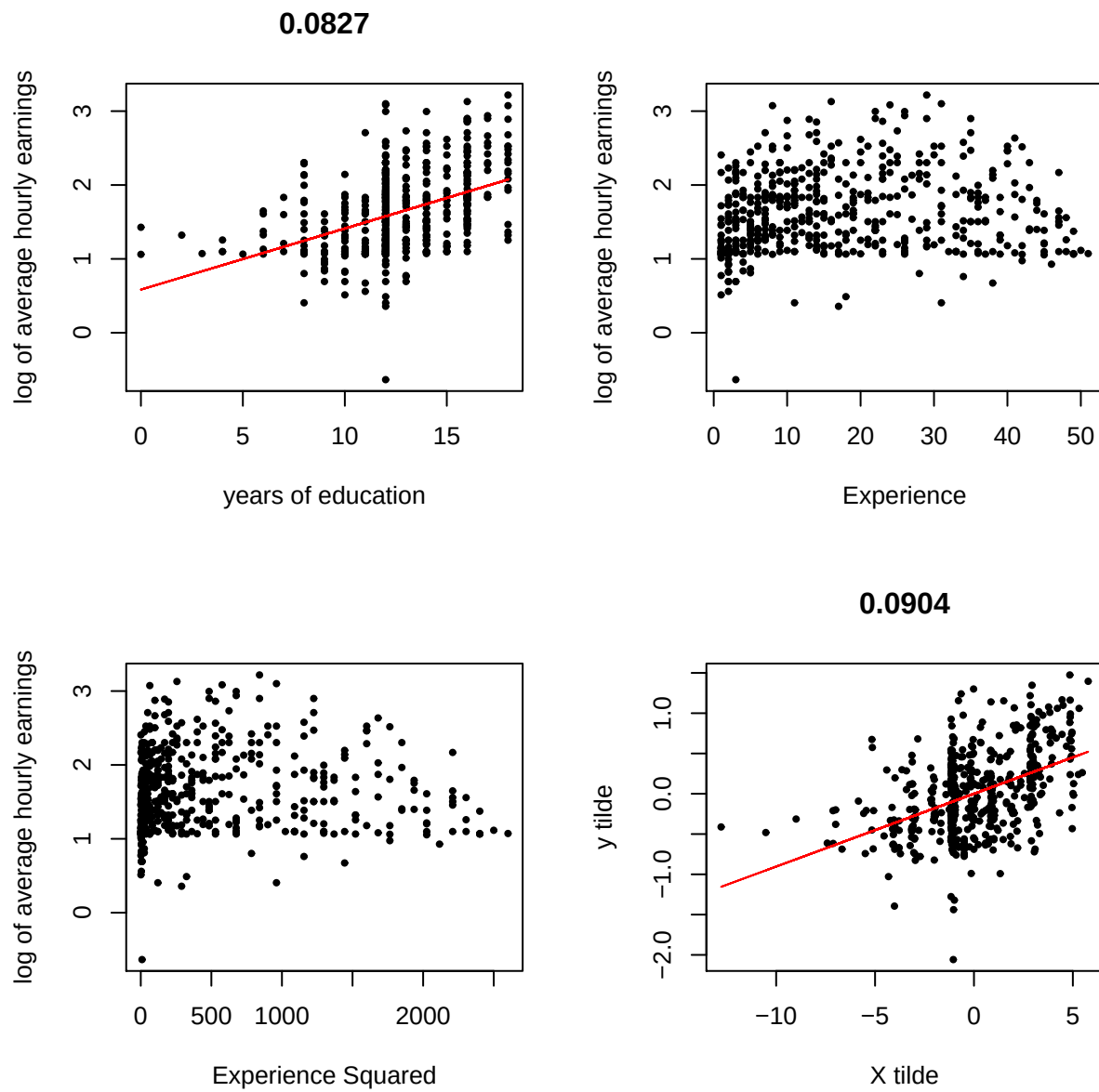


Figure 3: Partial Residual Plot

the residual in equation (14).

Example 3. Suppose we want to estimate the returns to education considering the following model,

$$lw_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i \quad (15)$$

where lw_i is the log of wages of the i^{th} worker, and x_{1i} , x_{2i} , and x_{3i} are the level of education, year of experience, and year with the actual employer of the i^{th} worker. The variable of interest is education, therefore we can rewrite the model using our previous notation:

$$y = Z'\gamma + \beta X + u$$

where $y = [lw]$, $Z = [x_2, x_3]$ and $X = [x_1]$.

We use a Wooldridge's (2002) data set to estimate the previous model. The data set contains observations for 500 individuals (Figure 1 shows the bivariate relation between dependent and independent variables). The last plot shows the slope obtained using partial residual plot. The return to education is 9 percent with an standard error of 0.0075.

5 Misspecification and Trade-offs

Under the classical assumptions the OLS estimator is unbiased, e.g. $E(\hat{\beta}) = \beta$. However, the estimator may be biased if we omit a variable in the regression model. Introducing variables in the model to 'correct' for the bias problem does not work because the variance of the estimators from the 'long' version of the model (e.g., the model that includes the additional explanatory variable) may be higher than the variance of the estimators from the 'short' version of the model (e.g., the model that does not include the explanatory variable).

Under the assumptions discussed so far, consider:

$$\begin{aligned} y_i &= x_i' \beta + z_i' \gamma + u_i \\ y_i &= x_i' b + v_i. \end{aligned}$$

Note that the second equation gives a projection of y on x_1 . In this case, we have:

$$\begin{aligned} b &= (Ex_i x_i')^{-1} (Ex_i y_i) = (Ex_i x_i')^{-1} (Ex_i (x_i' \beta + z_i' \gamma + u_i)) \\ &= (Ex_i x_i')^{-1} (Ex_i x_i') \beta + (Ex_i x_i')^{-1} (Ex_i z_i') \gamma + (Ex_i x_i')^{-1} (Ex_i u_i) \\ &= \beta + (Ex_i x_i')^{-1} (Ex_i z_i') \gamma, \end{aligned}$$

which obviously implies that the projection in the second equation does not identify a coefficient of interest β unless $Ex_i z_i' = 0$.

Let's discuss now the trade-off we often face in applied econometrics when we do not

observe a few independent variables. Suppose now that the true model is,

$$y = X\beta + Z\gamma + u, \quad (16)$$

but instead we estimate model,

$$y = X\beta + \epsilon. \quad (17)$$

The OLS estimator of the parameter of interest β ,

$$\begin{aligned} \hat{\beta}_s &= \arg \min \|y - X\beta\|^2, \\ &= (X'X)^{-1}X'y, \end{aligned}$$

could be biased if either $\gamma \neq 0$ or the observables independent variables X and Z are not independent,

$$E(\hat{\beta}_s) = \beta + E((X'X)^{-1}X'(Z\gamma + u)) = \beta + \Gamma\gamma$$

with $\Gamma = (X'X)^{-1}X'Z$. Note that Γ is the estimator of a regression of Z on X .

Example 4. We start studying the bias effect from omitting a relevant variable in the regression model. Consider that the true population model is,

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + u \quad (18)$$

but we obtain OLS estimators from a model like,

$$y = \beta_0 + \beta_1x_1 + u, \quad (19)$$

which excludes the relevant variable x_2 . Suppose we are interested in the slope parameter β_1 . In this case,

$$\begin{aligned} E(\tilde{\beta}_1) &= \beta_1 + \frac{\beta_2 \sum_{i=1}^n (x_{i1} - \bar{x}_1)x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2} \\ &= \beta_1 + \beta_2\tilde{\alpha}_1 \end{aligned}$$

What is the interpretation of $\tilde{\alpha}_1$? Note that if we have a simple regression model of the form,

$$x_2 = \alpha_0 + \alpha_1x_1 + \epsilon,$$

and we obtain the OLS estimator for the slope parameter α_1 , this is equal to

$$\tilde{\alpha}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

Therefore, the omitted variable bias is

$$E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\alpha}_1 \quad (20)$$

The bias is zero if either $\beta_2 = 0$, which means that the independent variable x_2 does not have an effect on the dependent variable y , or $\tilde{\alpha}_1 = 0$, which means that the independent variables x_1 and x_2 are uncorrelated. If $\beta_2 \approx 0$ or $\tilde{\alpha}_1 \approx 0$, the size of the bias would be small, and therefore it may not be a cause of concern.

Sometimes it is possible to infer the sign of the bias. If the sign of the unknown parameter β_2 is equal to the sign of the correlation between the independent variables x_1 and x_2 , the bias is positive. In this case, we say that $\hat{\beta}_1$ is upward bias. On the other hand, if the sign of β_2 is different than the sign of the correlation between the independent variables, the bias is negative. It is said that in this case, the estimator $\tilde{\alpha}_1$ is downward bias.

Example 5. Consider that y is log of wages, x_1 denotes years of education, and x_2 is ability or ‘spunk’ of some workers. We may believe that β_2 is positive and different from zero because there are evidence that suggests that more talented people receive higher wages. On the other hand, we suspect that $\text{Corr}(x_1, x_2) \neq 0$. The talented people tends to spend more years in school, so we infer that the relationship between education and ability is positive. Consequently the bias may be positive, so it may be the case that the OLS estimate of β_1 .

Although $\hat{\beta}_s$ is biased, it is efficient compared to the estimator of β_1 from the long model,

$$\hat{\beta}_l = (X' M_z X)^{-1} (X' M_z y)$$

To see this, first write the estimator from a short model as,

$$\hat{\beta}_s = (X' X)^{-1} X' (\hat{y}_l + \hat{u})$$

where $\hat{y}$ and $\hat{u}$ are the vectors of fitted values and residuals from the long model. Thus,

$$\hat{\beta}_s = \hat{\beta}_l + \Gamma \hat{\gamma}$$

Thus the variance of the estimator in the long model is

$$V(\hat{\beta}_l) = V(\hat{\beta}_s - \Gamma \hat{\gamma}) = V(\hat{\beta}_s) + \Gamma V(\hat{\gamma}) \Gamma', \quad (21)$$

which implies that

$$V(\hat{\beta}_l) \geq V(\hat{\beta}_s),$$

since the quadratic term is positive definite. The result in equation (21) holds if the (missing) covariance term is zero. We verify the conditions under which equation (21) holds in the following theorem:

Theorem 3. $\text{Cov}(\hat{\beta}_s, \Gamma \hat{\gamma}) = 0$

Proof: First, recall that

$$\begin{aligned}\hat{\gamma} &= (Z'M_xZ)^{-1}(Z'M_xy) \\ &= \gamma + (Z'M_xZ)^{-1}(Z'M_xu).\end{aligned}$$

Also, recall that

$$\begin{aligned}\hat{\beta} &= (X'M_zX)^{-1}(X'M_zy) \\ &= \beta + \Gamma\gamma + (X'M_zX)^{-1}(X'M_zu).\end{aligned}$$

By definition, we have that

$$\begin{aligned}Cov(\hat{\beta}_s, \Gamma\hat{\gamma}) &= E \left\{ \left(\hat{\beta}_s - E(\beta_s) \right) (\hat{\gamma} - E(\hat{\gamma}))' \Gamma' \right\} \\ &= E \left\{ \left(\hat{\beta}_s - E(\beta_s) \right) (\hat{\gamma} - \gamma)' \Gamma' \right\} \\ &= E \left\{ (X'X)^{-1} X'u ((Z'M_xZ)^{-1} ZM_xu)' \Gamma' \right\},\end{aligned}$$

by replacing by definitions. The last expression can be written as

$$\begin{aligned}Cov(\hat{\beta}_s, \Gamma\hat{\gamma}) &= E \left((X'X)^{-1} X'uu' M_xZ (Z'M_xZ)^{-1} \Gamma' \right) \\ &= \sigma^2 \left((X'X)^{-1} X' M_xZ (Z'M_xZ)^{-1} \Gamma' \right)\end{aligned}$$

because $V(u) = \sigma^2 I$. Note that $M_xX = 0$, therefore, covariance is equal to zero under homocedasticity. ■

We now summarize the main points. Since in applied economics usually both (a) the independent variables are correlated, and (b) there are variables that are omitted since there are difficult to measure,

1. Omitting independent variables introduce bias,
2. Adding independent variables inflate variance.

Remark 2. Suppose now that we are interested on the effect of omitting a variable that is correlated with X_1 . Suppose now that there is matrix of variables Z and, consider decomposing X as,

$$X\beta = X_1\beta_1 + X_2\beta_2 = P_zX_1\beta_1 + M_zX_1\beta_1 + X_2\beta_2$$

For $\gamma = 0$, we have that

$$\begin{aligned}X\beta &= P_zX_1(\beta_1 + \gamma) + M_zX_1\beta_1 + X_2\beta_2 \\ &= X_1\beta_1 + X_2\beta_2 + P_zX_1\gamma\end{aligned}$$

Therefore, we can now use an F test on $H_0 : \gamma = 0$ using the following regression

$$y = X\beta + \hat{X}_1\gamma + u$$

This is related to the auxiliary regression version of the celebrated Hausman (1978) test (see Wu (1973) also). Note that if you replace $\hat{X}_1$ by a matrix containing powers of $\hat{y}$ (e.g., $\hat{y}^2$, $\hat{y}^3$, etc.), we have Ramsey's test for specification error.

6 Practical Implications

If X and Z in equation (16) are not correlated, then there is no omitted variable bias and, if the PRL is correctly specified, the parameter β_1 in (17) can have a causal interpretation. Experiments however are not common, so researchers rely on control variables to eliminate biases.

This is connected to what is known as the *conditional independence assumption*, or CIA (as in Angrist and Pischke 2009). This is a condition we want to satisfy for estimation and inference and here is why. Consider a simple bivariate model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where β_1 is a constant causal parameter or effect. For instance, y_i is earnings and x_i is having a college degree or an indicator of veteran status. The error term ϵ_i capture other factors affecting earnings. If,

$$\epsilon_i = \delta_0 + \delta_1 z_i + v_i,$$

where z_i denote an “observable” variable or simply a vector of observables. Because v_i is assumed to not be correlated with z_i , we have that $E(\epsilon_i|x_i) \neq 0$ but $E(v_i|x_i, z_i) = 0$. Therefore, we estimate an “augmented” model with control z_i ,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + u_i,$$

by OLS, obtaining an unbiased estimate of β_1 . Note that this was not possible if we use OLS to estimate (6).

Example 6. The role of covariates can be seen in the illustration presented in Table 1. We use Social Security and administrative data from Angrist (1998) which studied the effect of voluntary military service on earnings of soldiers. We use workers who applied to the military between 1976 and 1981, and we estimate the model before military service and after (i.e., years 1974 and 1985, as shown in the table). Voluntary military service is obviously non-random and most likely correlated with education, age, test scores, cohort effects, etc. We also know that these observables affect earnings. Therefore, omission of these variables

Table 1: The effect of veteran status on life-cycle earnings by race

	Log earnings (non-white)				Log earnings (white)			
	74	74	85	85	74	74	85	85
Veteran	0.002 (0.074)	-0.034+ (0.018)	0.131** (0.012)	0.085** (0.007)	-0.079 (0.072)	-0.049** (0.018)	0.040** (0.010)	-0.003 (0.006)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Observations	1,763	1,763	2,499	2,499	2,096	2,096	3,155	3,155
R-squared	0.000	0.942	0.048	0.688	0.001	0.942	0.005	0.664
N	1763	1763	2499	2499	2096	2096	3155	3155

Standard errors in parentheses

** p<0.01, * p<0.05, + p<0.1

are likely to introduce biases.

The point is that after controlling for all these “observable” differences (using interactions too), veterans and non-veterans are comparable. Whether you believe or not this premise, the results shown in Table 1 points to the fact that the model estimated without controls give results that appear misleading.

Remark 3. We refer to controls as exogenous variables or observables, not to variables that are determined by the variable of interest. These controls are not potentially right hand side variables for other equations (or endogenous variables in genera). For instance, in the Mincer equation,

$$\log(wages)_i = \beta_0 + \beta_1 educ_i + \alpha_i + u_i,$$

α_i measures ability of the worker i and it is not observed (part of the error term). It makes sense to assume that $\text{Cov}(\alpha_i, educ_i) \neq 0$. In the CIA world, it is tempting to include variables to satisfy the condition and, sometimes, these variables are available to practitioners. For instance, you might have test scores used to screen job applicants. The real problem is that the performance in the test is likely to be correlated with education too.

For simplicity, consider that test scores are modeled as $score_i = \pi_0 + \pi_1 educ_i + \alpha_i + \varepsilon_i$. If we use test scores as a control variable, we estimate

$$\log(wages)_i = \gamma_0 + \gamma_1 educ_i + \gamma_2 score_i + v_i,$$

where $\gamma_1 = \beta_1 - \pi_1 \neq \beta_1$. Angrist and Pischke (2009) call these variables, including the test score used in this example, ‘bad controls’.